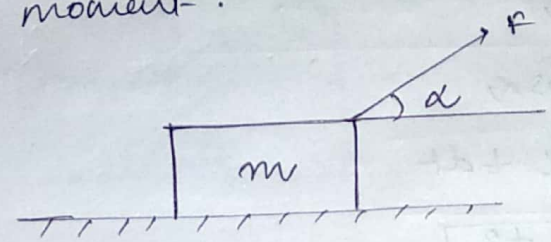
Tutorial-3

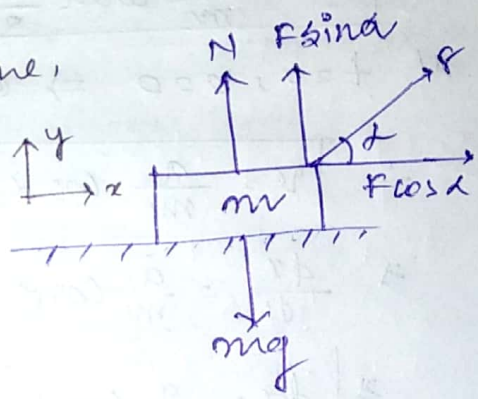
12/09/20

5. At the moment $t=0$, the force $F=at$ is applied to a small body of mass m resting on a smooth horizontal plane (a is a const). The permanent direction of this force forms an angle α with the horizontal. Find (a) the velocity of the body at the moment of its breaking off the plane, (b) the distance traversed by the body upto this moment.

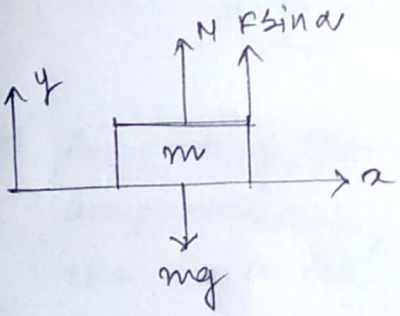
⇒



⇒ as smooth horizontal plane, so, no frictional force is there.



Consider the y-component of forces →



N → reaction of force (contact force)

$$N + F\sin\alpha = mg$$

at the time of breaking off $N=0$

$$\Rightarrow F\sin\alpha = mg$$

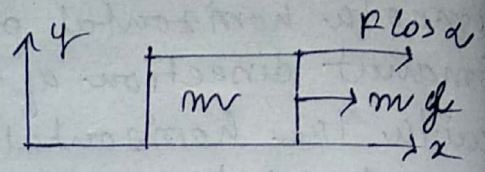
$$\Rightarrow at\sin\alpha = mg$$

$$\Rightarrow \boxed{t = \frac{mg}{a\sin\alpha}} \quad \text{(1)}$$

So, we get the time, of t at moment when the block will just break off from the surface.

Consider the x component of forces –

If 'f' is the acc. then



$F\cos\alpha = mf$

$\cancel{\Rightarrow m\frac{dv}{dt}\cos\alpha =}$

$\Rightarrow at\cos\alpha = mf$

$\Rightarrow mf = at\cos\alpha$

$\Rightarrow m\frac{dv}{dt} = at\cos\alpha$

$\Rightarrow \int dv = \int \frac{a}{m}\cos\alpha\, t\, dt$

$\Rightarrow \boxed{v = \frac{a}{m}\cos\alpha\frac{t^2}{2}} + c_1$

at $t=0, v=0 \Rightarrow c_1 = 0$

$\therefore \boxed{v = \frac{a}{m}\cos\alpha\frac{t^2}{2}}$ —(2)

$\Rightarrow \frac{dx}{dt} = \frac{a}{m}\cos\alpha\frac{t^2}{2}$

$\Rightarrow \int dx = \frac{a}{m}\cos\alpha\int\frac{t^2}{2}dt$

$\Rightarrow x = \frac{a}{m}\cos\alpha\cdot\frac{1}{6}t^3 + c_2$

at $t=0, x=0 \Rightarrow c_2 = 0$

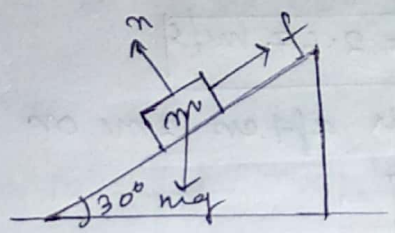
$\therefore \boxed{x = \frac{a}{6m}\cos\alpha\, t^3}$ —(3)

So, putting $t = \frac{mg}{a\sin\alpha}$ [from (1)] into (2) & (3), →

$$\boxed{\begin{aligned} v &= \frac{mg^2}{2a}\frac{\cos\alpha}{\sin^2\alpha} \\ s &= \frac{m^2g^3}{6a^2}\frac{\cos\alpha}{\sin^3\alpha} \end{aligned}}$$

Scanned with CamScanner

3. A 3 kg block, as shown in figure, starts from rest at the top of a 30° inclined and slides a distance of 2m down the incline in 1.5 s. find (a) the magnitude of the acc. of the block. (b) the co-efficient of kinetic friction between the block and plane (μ) (c) the frictional force acting on the block and (d) the speed of the block after it has slide 2m.



⇒ (a) we know, $s = ut + \frac{1}{2}at^2$ (for const acceleration)

Now, as the block starts from rest,

$u = 0$

$\therefore s = \frac{1}{2}at^2$

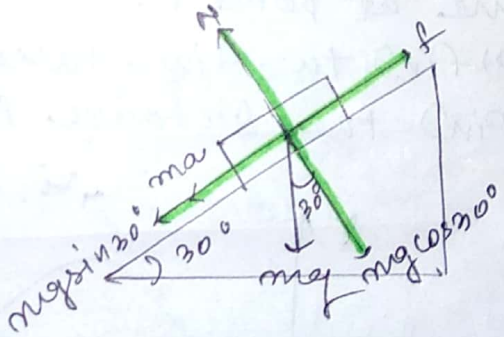
$\Rightarrow a = \frac{2s}{t^2} = \frac{2 \times 2m}{(1.5)^2 s^2} = 1.78 \, m/s^2$

$\Rightarrow \boxed{a = 1.78 \, m/s^2}$

(b) Comparing the force components,

$N = mg\cos 30°$

$= 3 \times 9.8 \times \frac{\sqrt{3}}{2}$



Now if, $\mu \rightarrow$ co-efficient of kinetic friction, then,

$f = \mu N$

$= \mu mg\cos 30°$

$\Rightarrow \boxed{\mu = \frac{f}{mg\cos 30°}}$

$\Rightarrow \mu = \frac{9.36}{3 \times 9.8 \times \sqrt{3}/2}$

$\boxed{\mu = 0.368}$

from figure,

$mg\sin 30° - f = ma$

$\Rightarrow f = mg\sin 30° - ma$

$= m[g\sin 30° - a]$

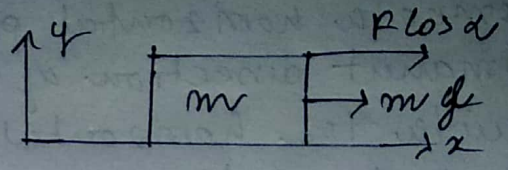
$= 3[9.8 \times \frac{1}{2} - 1.78]$

$\boxed{f = 9.36 \, N}$

Scanned with CamScanner

Consider the x component of forces –

If 'f' is the acc, then



$F\cos\alpha = mf$

$\cancel{\Rightarrow m\frac{dv}{dt}\cos\alpha =}$

$\Rightarrow at\cos\alpha = mf$

$\Rightarrow mf = at\cos\alpha$

$\Rightarrow m\frac{dv}{dt} = at\cos\alpha$

$\Rightarrow \int dv = \int \frac{a}{m}\cos\alpha\, t\, dt$

$\Rightarrow \boxed{v = \frac{a}{m}\cos\alpha\frac{t^2}{2}} + c_1$

at $t=0$, $v=0 \Rightarrow c_1 = 0$

$\therefore \boxed{v = \frac{a}{m}\cos\alpha\frac{t^2}{2}}$ —(2)

$\Rightarrow \frac{dx}{dt} = \frac{a}{m}\cos\alpha\frac{t^2}{2}$

$\Rightarrow \int dx = \frac{a}{m}\cos\alpha\int\frac{t^2}{2}dt$

$\Rightarrow x = \frac{a}{m}\cos\alpha\cdot\frac{1}{6}t^3 + c_2$

at $t=0$, $x=0 \Rightarrow c_2 = 0$

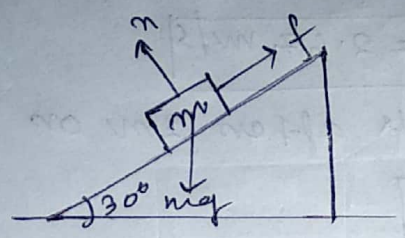
$\therefore \boxed{x = \frac{a}{6m}\cos\alpha\, t^3}$ —(3)

So, putting $t = \frac{mg}{a\sin\alpha}$ [from (1)] into (2) & (3), $\rightarrow$

$$\boxed{\begin{aligned} v &= \frac{mg^2}{2a}\frac{\cos\alpha}{\sin^2\alpha} \\ s &= \frac{m^2g^3}{6a^2}\frac{\cos\alpha}{\sin^3\alpha} \end{aligned}}$$

Scanned with CamScanner

3. A 3 kg block, as shown in figure, starts from rest at the top of a 30° inclined and slides a distance of 2 m down the incline in 1.5 s. Find (a) the magnitude of the acc. of the block. (b) the co-efficient of kinetic friction between the block and plane (μ) (c) the frictional force acting on the block and (d) the speed of the block after it has slide 2 m.



⇒ (a) we know, $s = ut + \frac{1}{2}at^2$ (for const acceleration)

Now, as the block starts from rest,

$u = 0$

$\therefore s = \frac{1}{2}at^2$

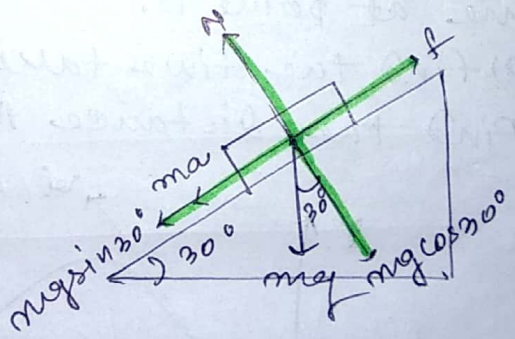
$\Rightarrow a = \frac{2s}{t^2} = \frac{2 \times 2\,m}{(1.5)^2 s^2} = 1.78\,m/s^2$

$\Rightarrow \boxed{a = 1.78\,m/s^2}$

(b) Comparing the force components,

$N = mg\cos 30°$

$= 3 \times 9.8 \times \frac{\sqrt{3}}{2}$



Now if, $\mu \rightarrow$ co-efficient of kinetic friction, then,

$f = \mu N$

$= \mu mg\cos 30°$

$\Rightarrow \boxed{\mu = \frac{f}{mg\cos 30°}}$

$\Rightarrow \mu = \frac{9.36}{3 \times 9.8 \times \sqrt{3}/2}$

$\boxed{\mu = 0.368}$

from figure,

$mg\sin 30° - f = ma$

$\Rightarrow f = mg\sin 30° - ma$

$= m[g\sin 30° - a]$

$= 3[9.8 \times \frac{1}{2} - 1.78]$

$\boxed{f = 9.36\,N}$

Scanned with CamScanner

(c) frictional force acting on the block $\boxed{f = 9.37\ N}$

(d) $v = u + at$

The block was initially at rest,

$u = 0$

$\therefore v = at$

$\Rightarrow \boxed{v = 1.78 \times 1.5 = 2.67\ m/s}$

So, the speed of block after 2m or 1.5s –

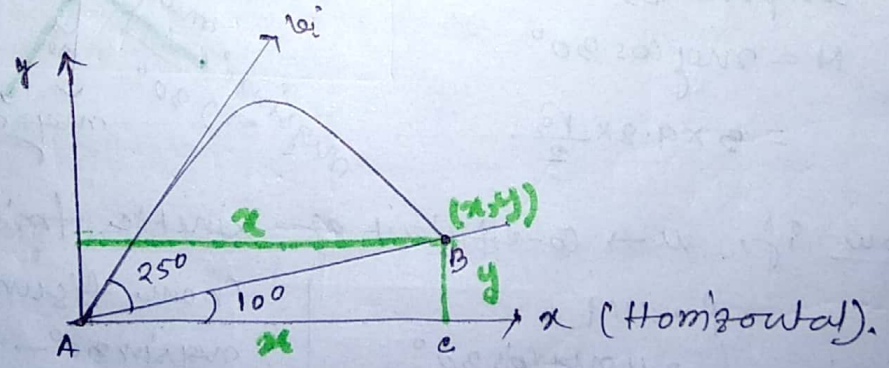
$\boxed{v = 2.67\ m/s}$

2. A projectile is thrown from a point 'A' at an angle of 25° with an inclined plane. The initial velocity of the ~~proj~~ projectile is – 20 m/s and the inclined plane makes an angle of 10° with the horizontal. After sometime, the projectile hits the inclined plane at point B.

(a) find the time taken to reach the point B.

(b) find the distance AB.

$\Rightarrow$



The projectile makes an angle of $(25° + 10°) = 35°$ with the horizontal.

Let, the co-ordinat of projectile at point $B \equiv (x, y)$ at any instant t.

Now, let the initial velocity = v_i = 20 m/s

So, $v_x = v_i \cos 35^\circ$ —— (1)

$v_y = v_i \sin 35^\circ$. —— (2)

So, $x = v_x \cdot t = v_i \cos 35^\circ t$ —— (3)

$y = v_y t - \frac{1}{2} g t^2$

$\Rightarrow y = v_i \sin 35^\circ t - \frac{1}{2} g t^2$ —— (4)

Now, from $\Delta ABC \rightarrow$

$$\tan 10^\circ = \frac{y}{x} = \frac{v_i \sin 35^\circ t - \frac{1}{2} g t^2}{v_i \cos 35^\circ \cdot t}$$

$$= \frac{v_i \sin 35^\circ - \frac{1}{2} g t}{v_i \cos 35^\circ}$$

$\Rightarrow 0.176 \times 20 \times 0.819 = 20 \times 0.573 - \frac{1}{2} \times 9.8 \times t$

$\Rightarrow \boxed{t = 1.75 \text{ sec}}$

(b) Now, $AB = \sqrt{AC^2 + BC^2}$

$= \sqrt{x^2 + y^2}$

$x = 20 \times \cos 35^\circ \times t$

$= 20 \times 0.819 \times 1.75$

$= 28.665$ m

$y = 20 \times 0.573 \times 1.75 - \frac{1}{2} \times 9.8 \times (1.75)^2$

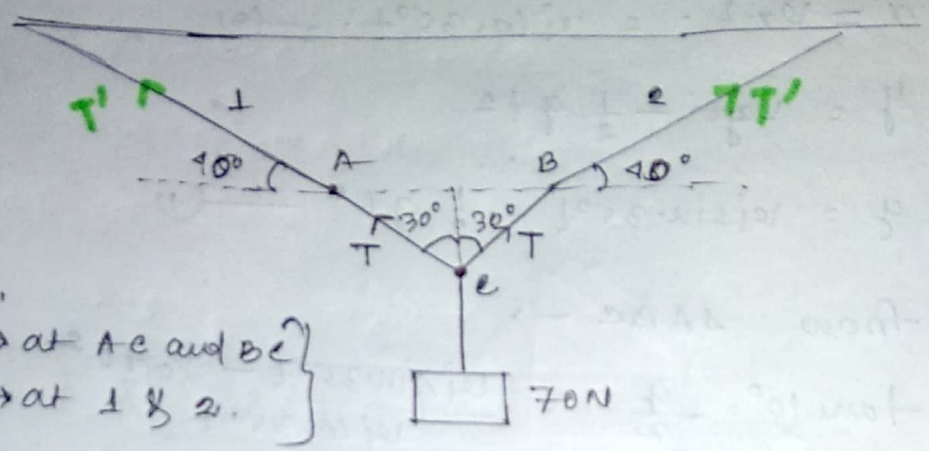
$= 5.048$ m

$\therefore AB = \sqrt{(28.665)^2 + (5.048)^2}$ m

$\boxed{AB = 29.11 \text{ m}}$

Scanned with CamScanner

1' A weight of 70N is suspended to a rigid support with the help of string as shown in the figure. Find the tension in the string AB.

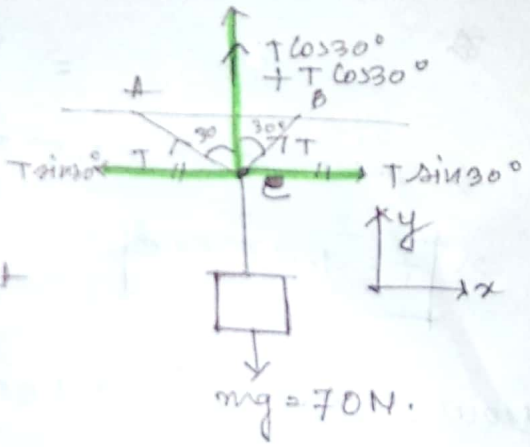


Let,
T → at AC and BC
T' → at 1 & 2.

⇒ At point 'c' —

The balancing force at x dirn is zero.

The balancing force at y dirn



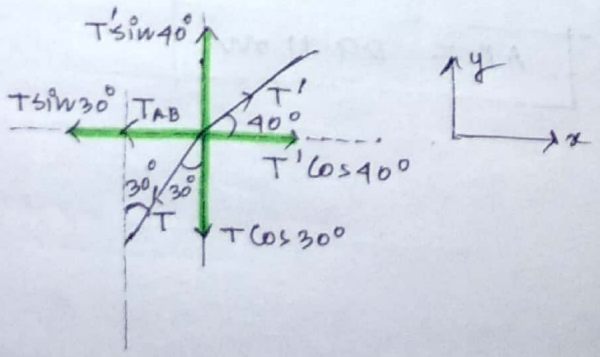
~~T30°+~~

$T\cos 30° + T\cos 30° = mg = 70$

$\Rightarrow 2T \cdot \frac{\sqrt{3}}{2} = 70$

$\Rightarrow \boxed{T = 70/\sqrt{3} = 40.4\,N}$ —(1)

Now, we have the free-body diagram at point 'B' →



Fx →

$T\sin 30° + T_{AB} = T'\cos 40°$ —(2)

Fy' →

$T'\sin 40° = T\cos 30°$

$\Rightarrow T' = T\frac{\cos 30°}{\sin 40°}$

$= 40.4 \times \frac{\sqrt{3}}{2} \times \frac{1}{0.64} \Rightarrow \boxed{T' = 54.4\,N}$ —(3)

Scanned with CamScanner

from (2) →

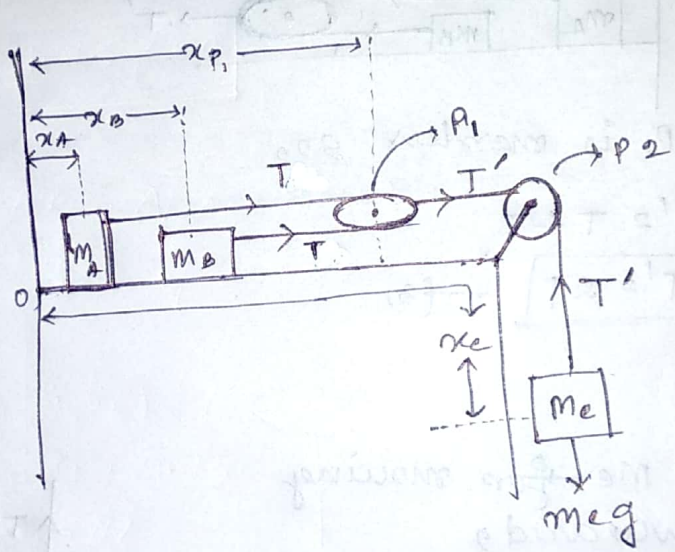
~~T sin 30° =~~ $T'\cos 40° - T\sin 30° = T_{AB}$

$\Rightarrow T_{AB} = 54.4\cos 40° - 40.4\sin 30°$

$\Rightarrow \boxed{T_{AB} = 21.47\ N}$

1. Two masses m_A & m_B, lies on a frictionless table, as shown, they are attached to either end of a light rope of length 'l' which passes around the pulley of negligible mass. the pulley is attached to a rope connected to a hanging mass, M_e. Find the acc. of each mass.

⇒



Consider the rope to be inextensible

Now, if we move m_e by a distance x then P_1 will also move by a distance x.

so, the distance betn P_1 and M_e always remains same.

i.e $x_e - x_{P_1} = \text{const}$.

diff. w.r.to 't', two times —

$\ddot{x}_e - \ddot{x}_{P_1} = 0$

$\Rightarrow \boxed{\ddot{x}_e = \ddot{x}_{P_1}}$ — (1) i.e the acc. of P_1 and M_e is same.

Scanned with CamScanner

Now, also the displacement of P_1, the distance between P_1 and m_A & P_1 and m_B remain Const.

i.e,

$$(x_{P_1} - x_A) + (x_{P_1} - x_B) = \text{Const}$$

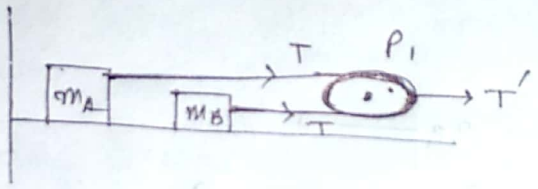
$$\Rightarrow 2x_{P_1} - (x_A + x_B) = \text{Const}$$

diff. w.r.to 't', two times –

$$\Rightarrow 2\ddot{x}_{P_1} = \ddot{x}_A + \ddot{x}_B$$

$$\Rightarrow \boxed{\ddot{x}_A + \ddot{x}_B = 2\ddot{x}_{P_1} = 2\ddot{x}_e} \quad \text{—(2)}$$

Now



As, the P_1 is massless so,

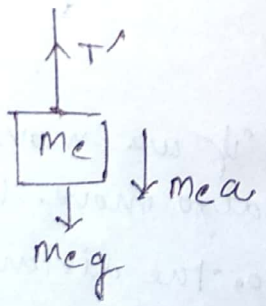
$$T' = T + T$$

$$\Rightarrow \boxed{T' = 2T} \quad \text{—(3)}$$

Now, for m_e for moving downward,

$$m_e g - T' = m_e a$$

$$\Rightarrow \boxed{m_e g - T' = m_e \ddot{x}_e} \quad \text{—(4)}$$



for m_A →

$$m_A \ddot{x}_A = T \Rightarrow \boxed{\ddot{x}_A = T/m_A} \quad \text{—(5)}$$

for m_B → $m_B \ddot{x}_B = T \Rightarrow \boxed{\ddot{x}_B = T/m_B}$ —(6)

Scanned with CamScanner

from (2) $\longrightarrow$

$$\frac{T}{m_A} + \frac{T}{m_B} = 2\left(g - \frac{T'}{m_C}\right)$$

$$= 2\left(g - \frac{2T}{m_C}\right) \quad [\text{from } -(3)]$$

$$\Rightarrow T\left(\frac{1}{m_A} + \frac{1}{m_B} + \frac{4}{m_C}\right) = 2g$$

$$\Rightarrow T = \frac{2g \cdot m_A \cdot m_B \cdot m_C}{(4m_A m_B + m_B m_C + m_C m_A)}$$

so,

$$\left\{\begin{aligned} \ddot{x}_A &= \frac{2g \cdot m_B \cdot m_C}{(4m_A m_B + m_B m_C + m_C m_A)} \\ \ddot{x}_B &= \frac{2g \cdot m_A \cdot m_C}{(4m_A m_B + m_B m_C + m_C m_A)} \\ \ddot{x}_C &= g - \frac{2T}{m_C} \\ &= \frac{(m_A + m_B) m_C \cdot g}{(4m_A m_B + m_B m_C + m_C m_A)} \end{aligned}\right\}$$

Scanned with CamScanner